Paper Review • arXiv 2611.04870

PrimeSim

A Generalist World Model Simulator from Large-Scale Cross-Embodiment Demonstrations

Generalist cross-embodiment simulator

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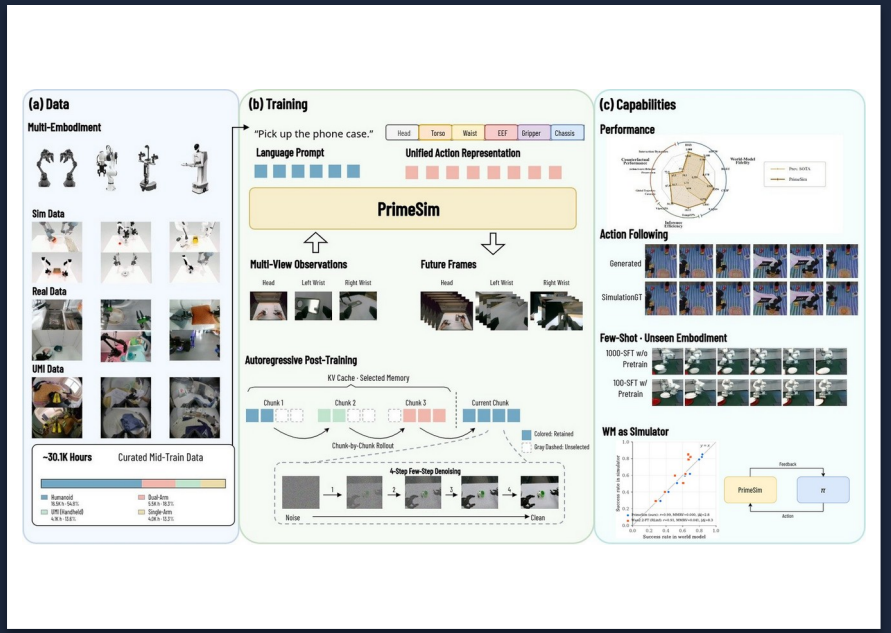
Submitted 2026-11-03 | Open: weights, inference & training code



Source: paper, Fig. 1

TL;DR: general video model → 10 FPS robot simulator for policy training

PrimeSim starts from Wan2.2-TI2V-5B and is mid-trained on **30,110 h of robot demos from 22 platforms**, learning **action-conditioned, three-view-consistent generation**; four-stage "fast video, slow memory" post-training then turns it into an autoregressive simulator with **4-step denoising, ~10 FPS and stable 60–105 s rollouts**. Its predicted policy success rates closely match a physics simulator, and RL can be run directly inside it.

30,110h	10.1FPS	66.6	6.09	ρ=1.00	61.7→73.2
From 22 platforms: mid-training corpus	4-step denoising Single A100	WorldSimProbe Mean; 1st on 4/5	EWMBench overall Ranked 1st	8-policy ranking matches sim. (r=0.99)	GRPO in PrimeSim, 200 steps Simulator success rate (%)

Contributions: large-scale cross-embodiment mid-training (unified action interface + frame-aligned action conditioning + view-aware attention) + "fast video, slow memory" post-training (causal conversion, 4-step distillation, self-rollout DMD, hierarchical memory).

Problem Definition

Video generation ≠ simulator: world models face three hard requirements

01 Action Following

Challenge

With global attention, each frame can aggregate actions from any timestep; existing models often "hallucinate" robot motion that contradicts the commands

PrimeSim's Solution

Unified 31-D action representation + first-frame-anchored relative action chunks + frame-aligned dual-path action conditioning

02 View Consistency

Challenge

Wrist cameras move with the arms and see each other; independent views become inconsistent, but free global attention drowns view-specific info

PrimeSim's Solution

View-aware interleaved attention (2 intra-view + 1 global block) + explicit camera-identity tokens

03 Long & Real-Time

Challenge

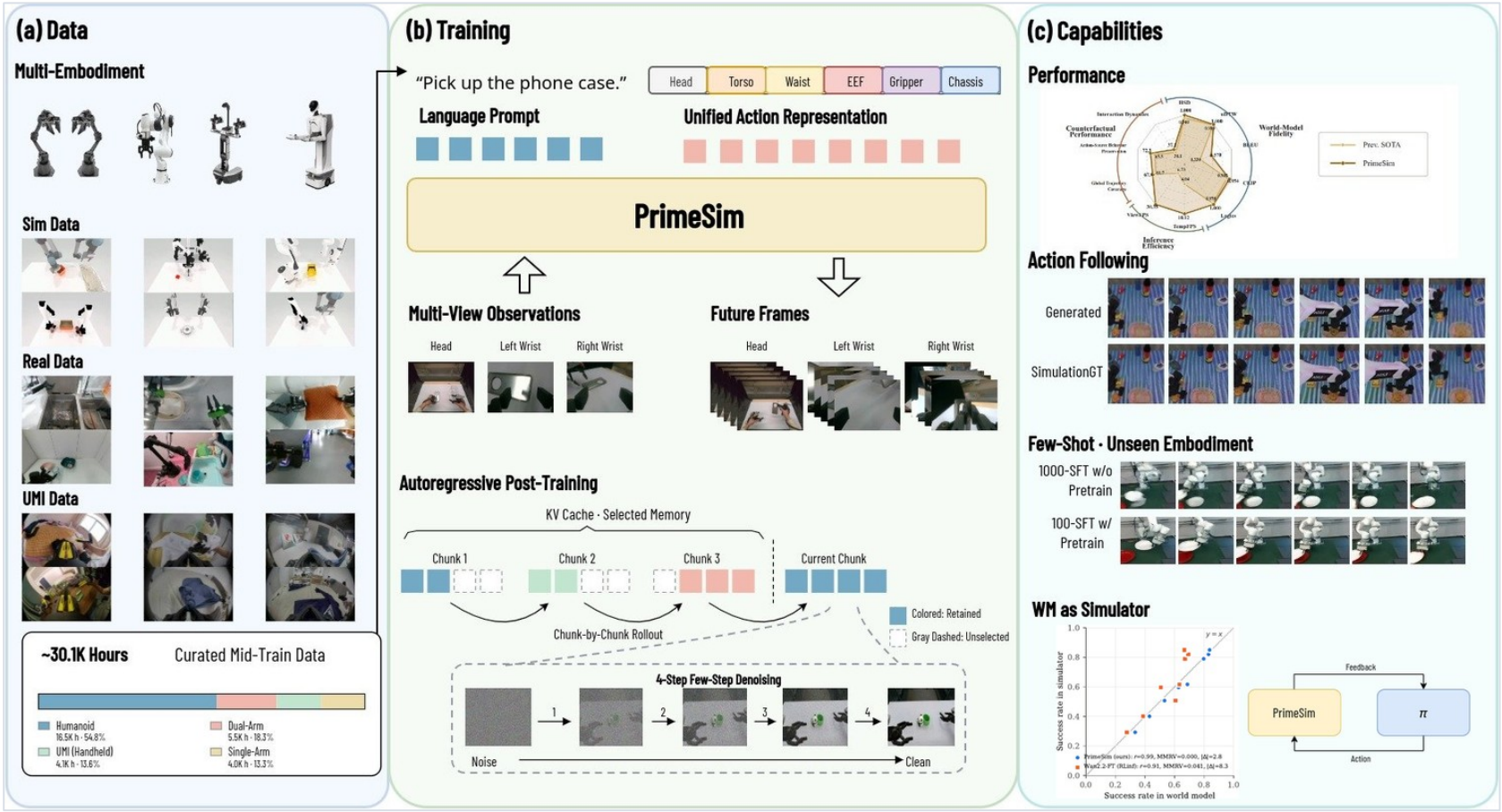
At inference the model conditions on its own imperfect predictions, so errors accumulate; bidirectional models need 50 denoising steps, which is too slow

PrimeSim's Solution

"Fast video, slow memory" post-training: causal chunking → 4-step distillation → self-rollout DMD → hierarchical temporal memory

Method Overview

Pre-train → mid-train → post-train: from video generator to simulator



Pre-training

Adopts Wan2.2-TI2V-5B (5.05B params) directly: generic video priors such as object permanence, lighting/materials and short-term motion

Mid-training

30,110 h • 22 platforms; unified actions + frame-aligned conditioning + view-aware attention; bidirectional flow matching

Post-training

Causal chunking → 4-step consistency distillation → self-rollout DMD → hierarchical memory; 10.1 FPS autoregressive simulation

Data: 30,110 h of robot-only demos, over half humanoid

30,110 hours

Total duration • 2.38B frames

1.52M

episodes (main text / Table 12)

22 platforms

20 dataset families • 1,773 repositories

84 : 16

Real : sim (by trajectory duration)

54.8%

Humanoid (16.5K h); dual-arm 18.3%

9,540 h

In-house humanoid chores, top single source

Note: Fig. 3 states 2.8M episodes and 85:15, inconsistent with the main text

Data Engine

4-stage data pipeline unifies heterogeneous sources into one format

Raw Sources

- Teleoperation
- UMI handheld gripper
- Simulation

① Video Quality

- Optical-flow motion scores control the distribution; near-static clips downsampled per platform
- Remove JPEG artifacts / numerical corruption
- Drop clips < 1 s

② Action Quality

- Remove dexterous hands and legged humanoids (keep wheeled ones)
- First-order-difference outlier detection for abrupt jumps
- Drop missing actions not recoverable via FK

③ Data Alignment

- Resample to 10 Hz (tolerance 0.03 s)
- Fixed 3 cameras: drop extras; if fewer, duplicate the main one
- Template prompts if no instruction; convert to LeRobot

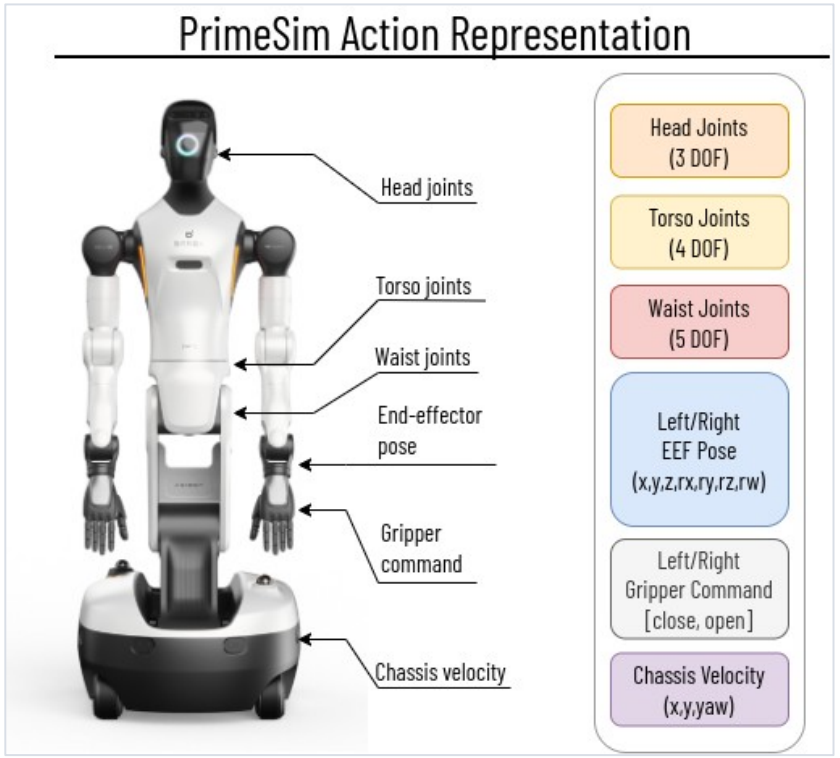
④ Action Alignment

- URDF + forward kinematics fill in EE poses
- Rotations unified as (x,y,z,w) quaternions
- 31-D unified representation + first-frame relative action chunks

Staged data use Mid-training uses the full corpus for coverage; post-training uses only a few complete, downstream-oriented episodes for long horizons and real-time speed

No human videos Pooled robot data now rivals egocentric human corpora in scale, avoiding latent-action / hand-to-end-effector alignment

Unified 31-D actions + relative chunks anchored at window frame 0



Head 3	Torso 4	Waist 5	EEF 2×7	Gripper 2	Base 3

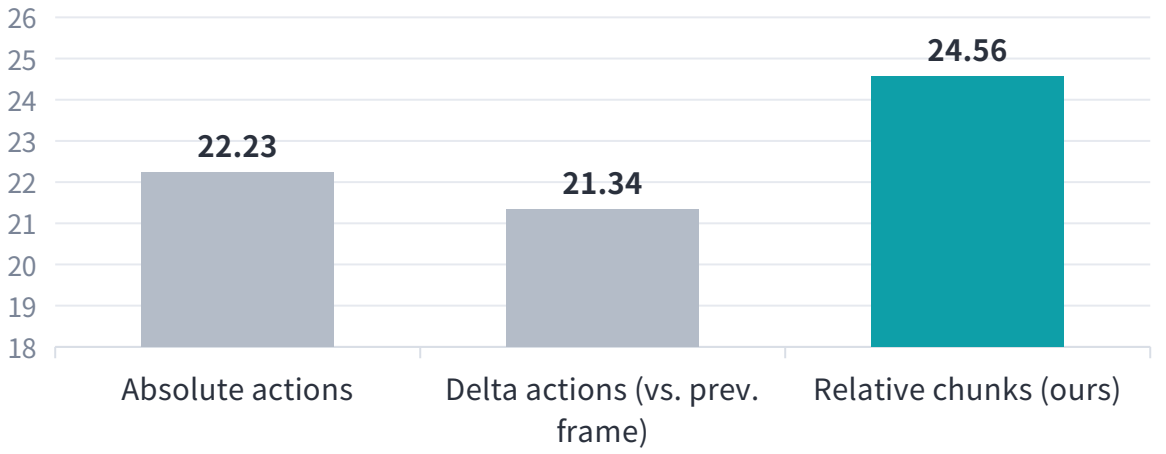
= 31-D; missing slots set to 0 (i.e., "no motion")

Relative chunks: all actions w.r.t. window frame 0

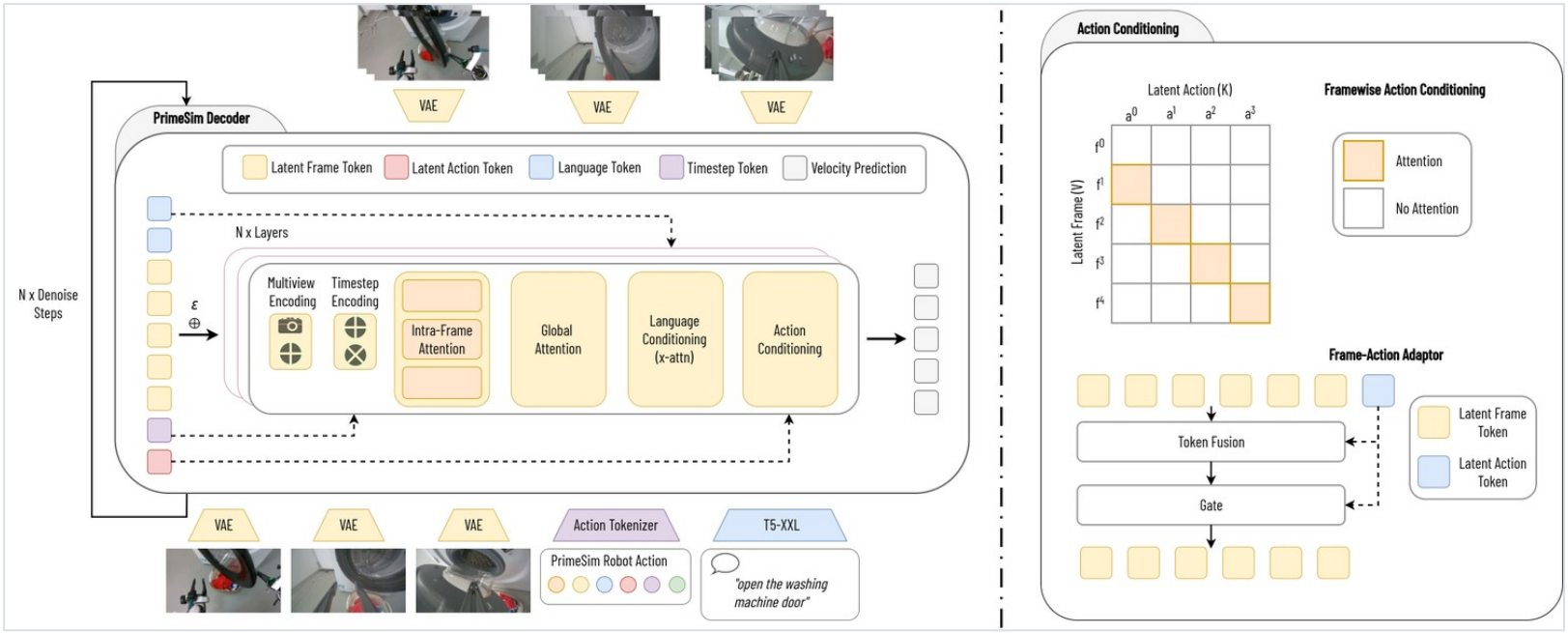
Joints: $\tilde{q}_t = q_t - q_0$ Poses: $\tilde{T}_t = T_0^{-1} T_t$ ($\tilde{p}_t = R_0^\top (p_t - p_0)$, $\tilde{R}_t = R_0^\top R_t$)
Gripper (binary) and base velocity (v_x, v_y, ω) unchanged

- Missing dims are naturally 0: no per-embodiment handling
- Removes each embodiment's absolute-origin offset
- All frames share one reference state: stronger, more concentrated conditioning (vs. per-frame deltas)

Table 9 • PSNR ↑ (y-axis not from 0)



Frame-aligned dual path: attention picks the action, Adapter enforces it



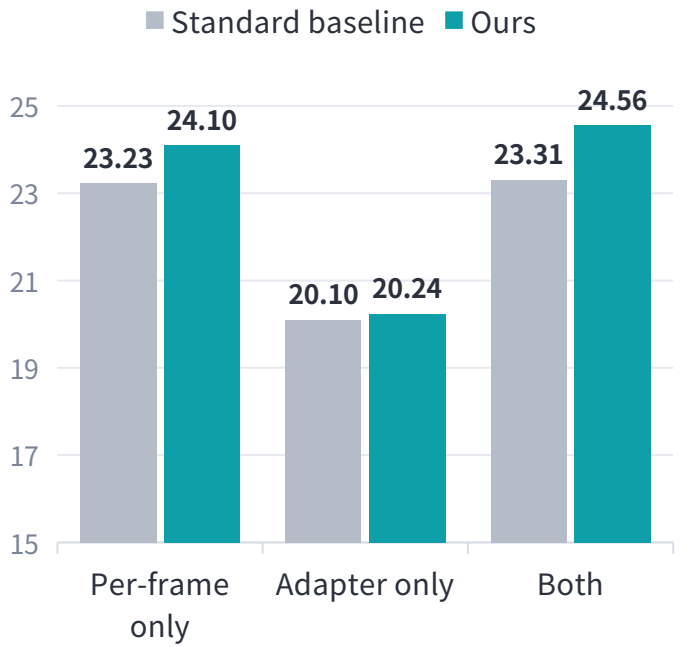


Table 10 • PSNR ↑ (baselines: cross-attn / AdaLN)

① Per-frame attn. (shifted diag. mask)

Action a_t drives the transition to frame t+1 → frame t attends only to a_{t-1}; the first frame gets no action, blocking interference from irrelevant timesteps at the attention level

② Action-Frame Adapter

The matching action token passes through Token Fusion + Gate to compute a residual feature offset injected directly into frame tokens, independent of learned attention weights

Action Tokenizer

A two-layer MLP encodes relative action chunks into action tokens, one per latent frame, giving both paths a shared time axis

View-aware interleaved attention: camera tokens give the biggest gain

Attention layout across 30 DiT blocks

Intra-view attn (2/3): keeps view-specific info

Global attn (every 3rd block): fuses shared scene

+ Learnable camera token prepended per view: identifies each token's source camera

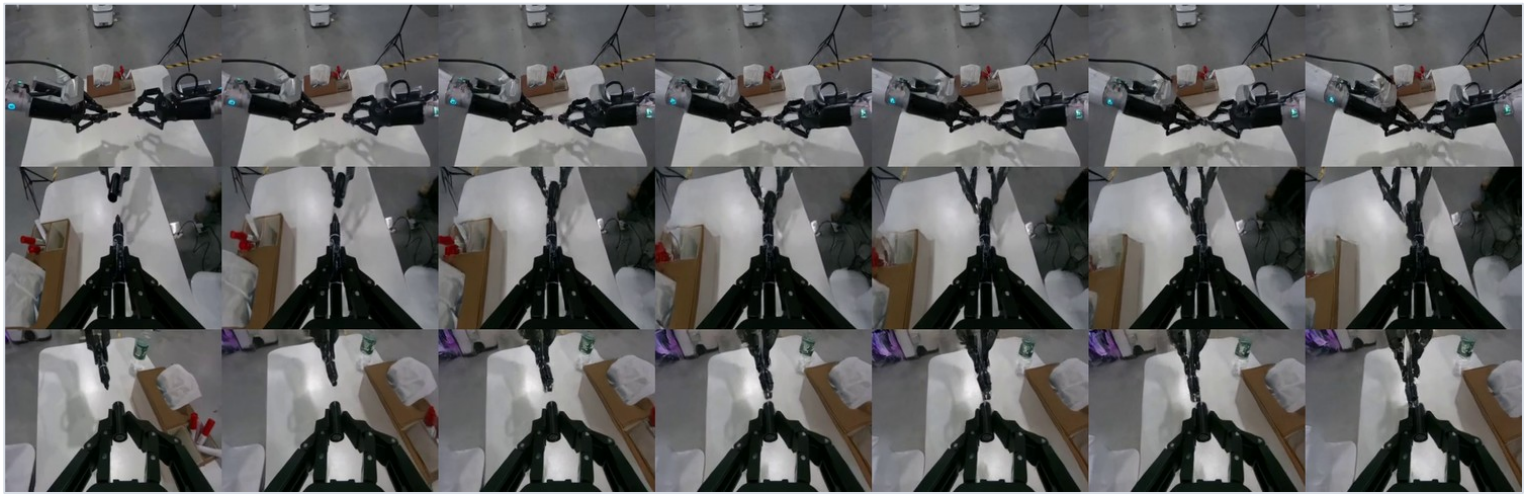


Fig. 11 Joint three-view generation (rows: head / left wrist / right wrist), driven by one action sequence

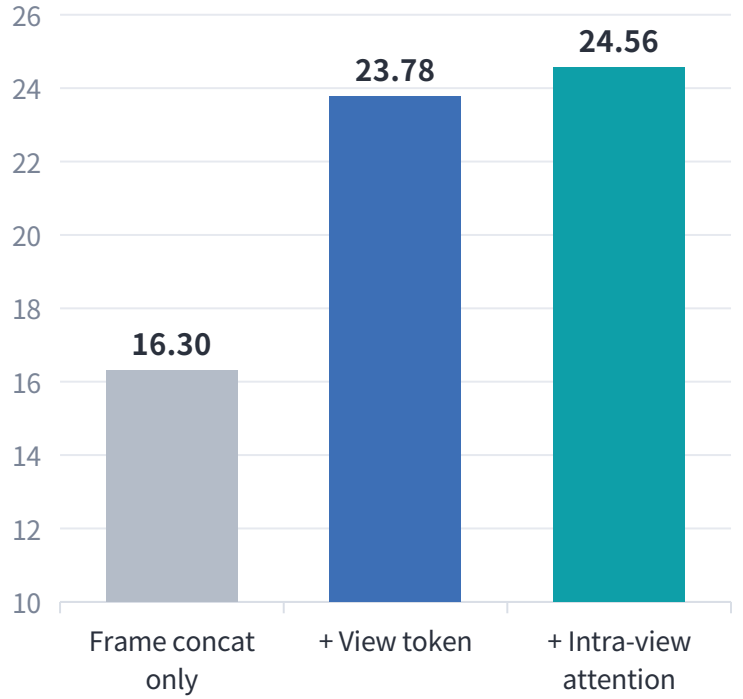


Table 11 • PSNR ↑ (components added incrementally)

Metric	Concat	+token	+Intra-view
SSIM ↑	0.62	0.83	**0.84**
LPIPS ↓	0.58	0.18	**0.15**

Post-training

Post-training in 4 stages: bidirectional generator → real-time simulator

Stage 1
Autoregressive Modeling

Method

Bidirectional within chunks, causal across chunks; teacher forcing (still flow matching, only the mask changes)

Solves

Enables chunk-by-chunk autoregressive generation

Stage 2
Causal Consistency Distillation

Method

Stage 1 model as frozen teacher; online matching of the teacher ODE's local transitions; LoRA-only training

Solves

50 → 4 steps, making self-rollout training affordable

Stage 3
Clip-Level Self-Rollout + Asymmetric DMD

Method

Student continues from its own KV cache; frozen bidirectional mid-trained model as teacher, reverse KL; fake-score net updated 5:1

Solves

Exposure bias; distillation-induced blur / averaging

Stage 4
Episode-Level Long-Horizon Adaptation

Method

Self-rollout extended to full episodes; hierarchical temporal memory; latent-space continuation; amortized supervision; actions anchored to episode start

Solves

Short-train/long-inference gap; forgetting after occlusion

50 → 4 Denoising steps	**≈ 9.0×** Stage 3 vs. Stage 1 throughput	**k = 3** 3 latent frames per chunk, 3 chunks per window	**r = 32** Stage 3 student LoRA rank

Same 21 slots: hierarchical memory degrades far less than a rolling cache

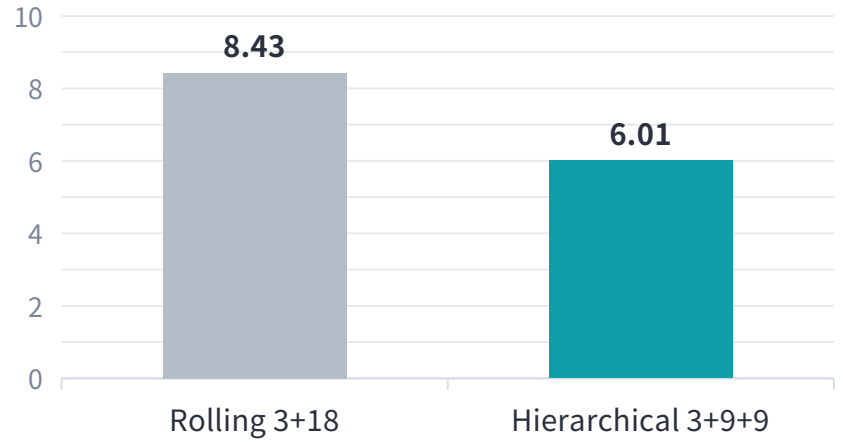
Hierarchical memory (3 + 9 + 9 = 21 slots)

Global sink | Slow: downsampled long-term | Local memory: dense recent motion

Baseline: rolling cache (3 + 18)

- After each window: old local memory merges into slow memory → temporally downsampled to fixed capacity, keeping time positions; frame indices synced across 3 views
- All but the initial frame are self-generated: same mechanism in training and inference, no teacher-forcing distribution gap
- Latent-space continuation without repeated decode/encode; actions anchored to episode start

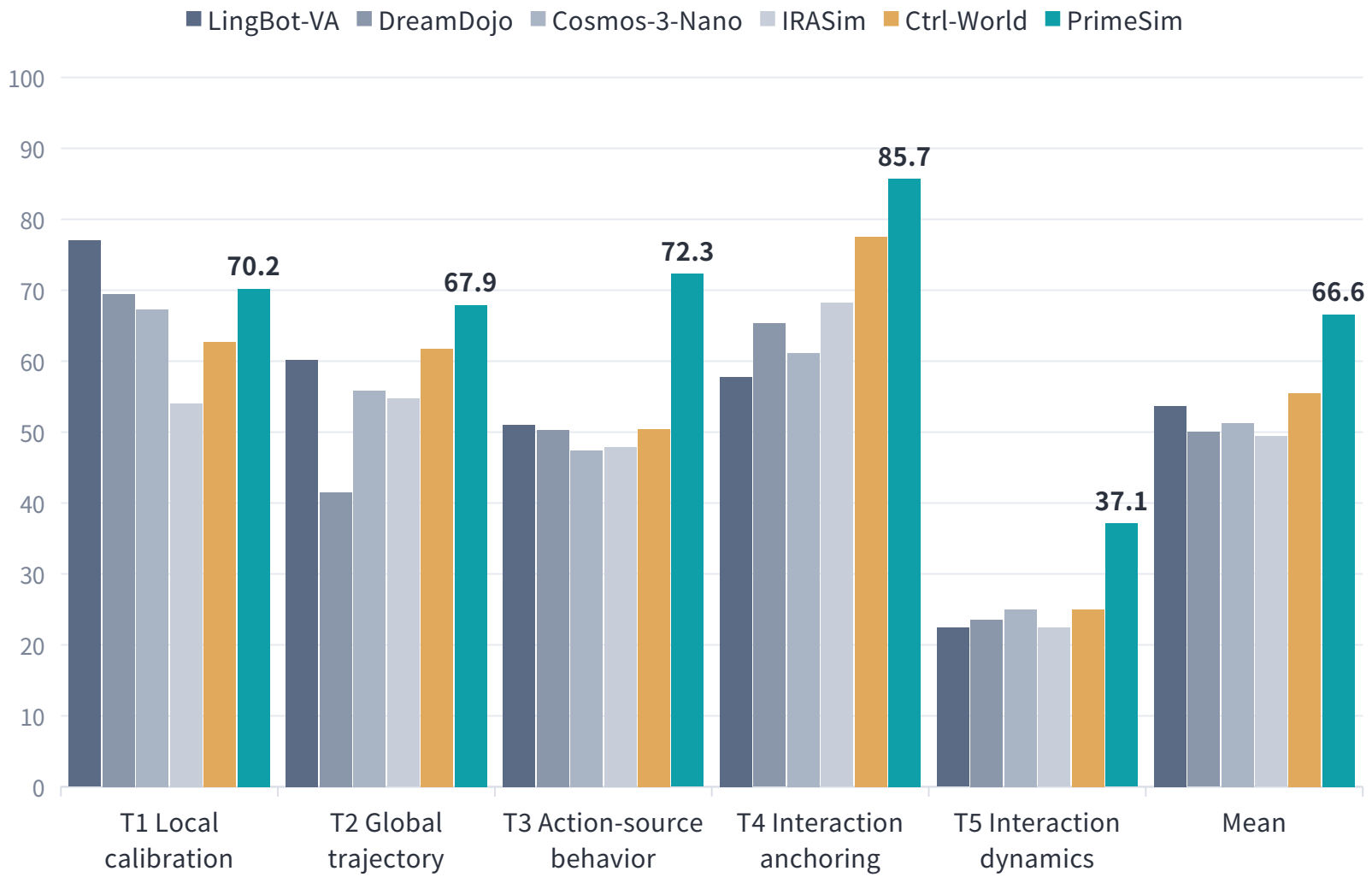
First-vs-last-window ΔPSNR ↓ (Table 8)



Decay ↓	Rolling	Hierarchical
ΔSSIM	0.120	**0.079**
ΔLPIPS	0.233	**0.159**

Fig. 12 RoboTwin 2.0 head view: arm fully hides lower-left shoe in frame 4; it reappears with the same pose and scale. Eval: 10 held-out eps., 61–105 s; GT: frame 1 only

WorldSimProbe: wins 4 of 5 subtasks, mean 66.6 leads by 11.1 points



What It Measures

"Observable simulator contract": action → corresponding motion → environment responds only to realized motion (failures PSNR cannot reveal)

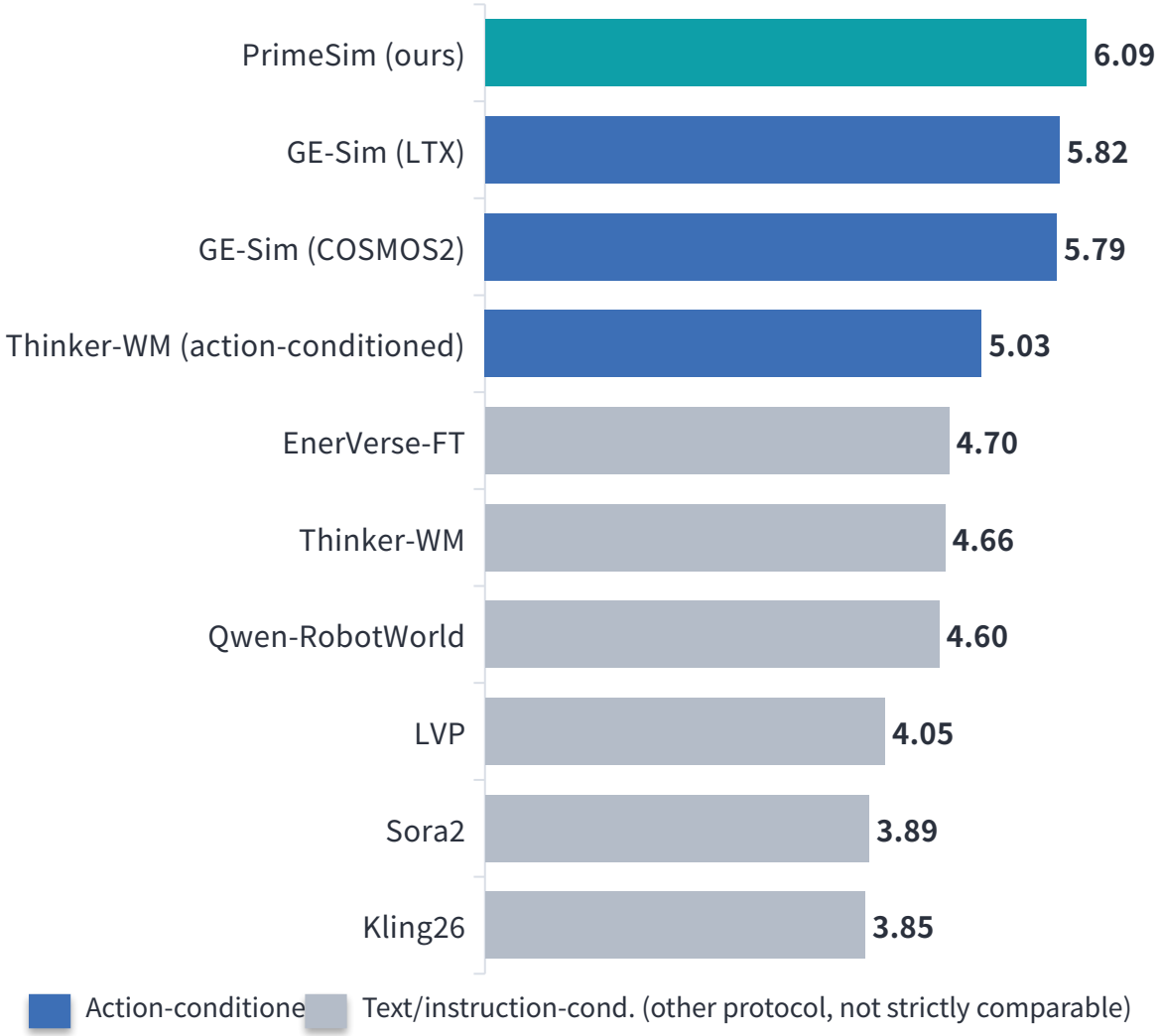
Leads 4/5: T2 67.9 (+10.0%), T3 72.3 (+41.8%), T4 85.7, T5 37.1 (+48.4%)

Only miss: T1 local calibration 70.2 < LingBot-VA 77.0; fine-grained control still not the best

T4 rationale: memory trained on its own rollouts learns how the environment responds to "actual motion"

Note: evaluation additionally built training data from VLA multi-trajectories / perturbed trajectories / swapped objects, plus visual noise; 192×256, 33 frames

EWMBench: top total 6.09, but measured after in-domain fine-tuning

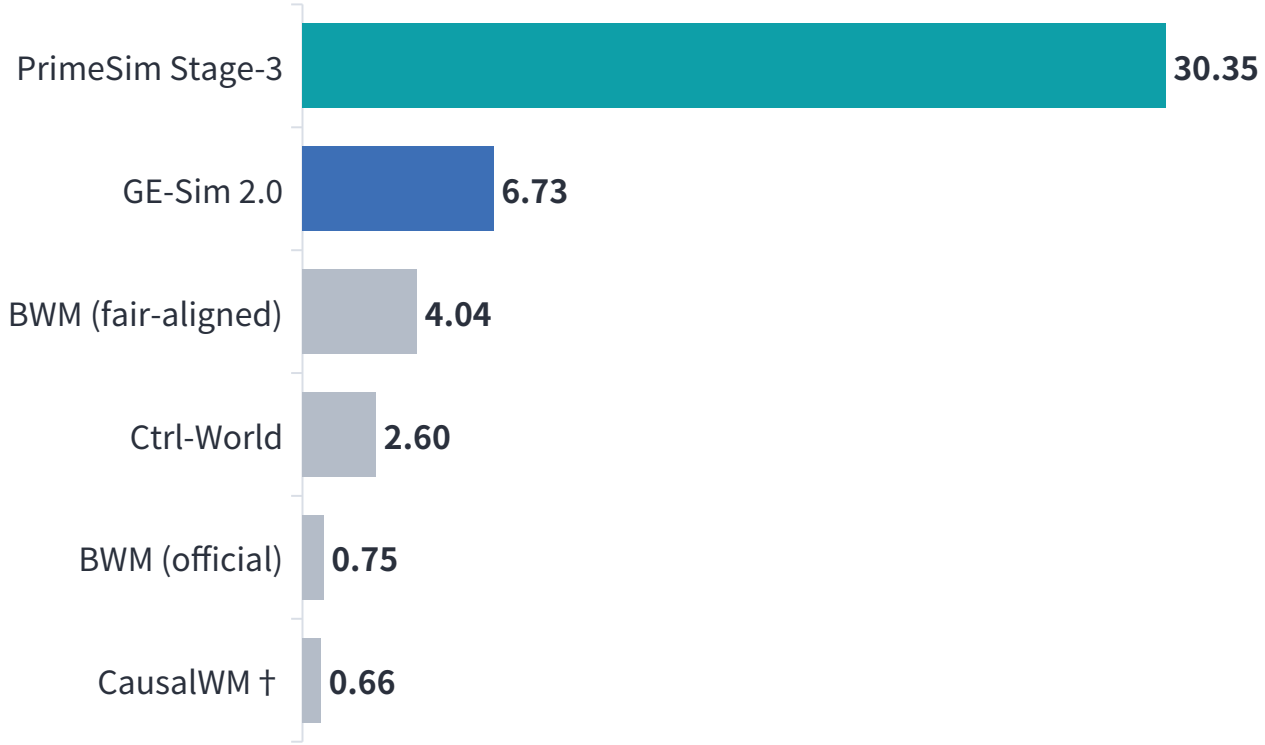


Per metric: PrimeSim vs. GE-Sim (LTX)

Dimension	PrimeSim	GE-Sim (LTX)
SceneC scene consistency	**0.887**	0.900
HSD	**1.000**	0.940
Dyn	**0.666**	0.780
nDTW trajectory alignment	**1.000**	0.980
Diversity	**0.010**	0.011
BLEU action semantics	**0.570**	0.330
CLIP	**0.954**	0.908
Logics logical consistency	**1.000**	0.970

Caveat AgiBot World is in the training corpus, and the model was fine-tuned on it before evaluation; uses 50 steps + CFG 1.5 (not the 4-step model) and explored a small family of sampling settings; several perfect scores may be saturated.

Speed: 30.35 ViewFPS, ~4.5× faster than GE-Sim 2.0 (also 4-step, 3-view)



Model	Views	Res./view	Steps	TempFPS
PrimeSim	3	288×384	4	**10.12**
GE-Sim 2.0	3	384×512	4	**2.24**
Ctrl-World	3	192×320	50	**0.87**

10.12
TempFPS
(timesteps/s)

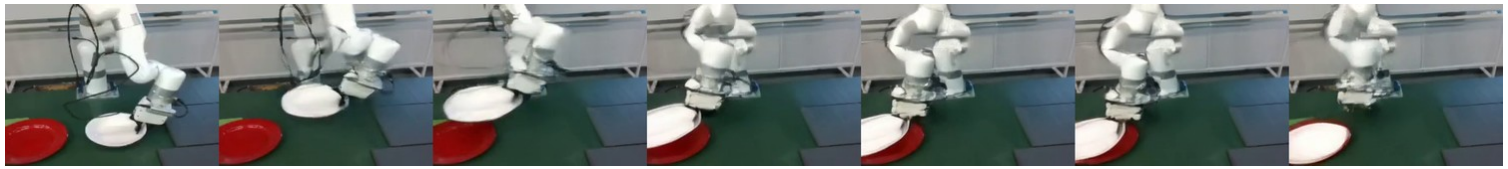
3.16 s
Per 33-frame
3-view window

Comparison caveats (per the paper)

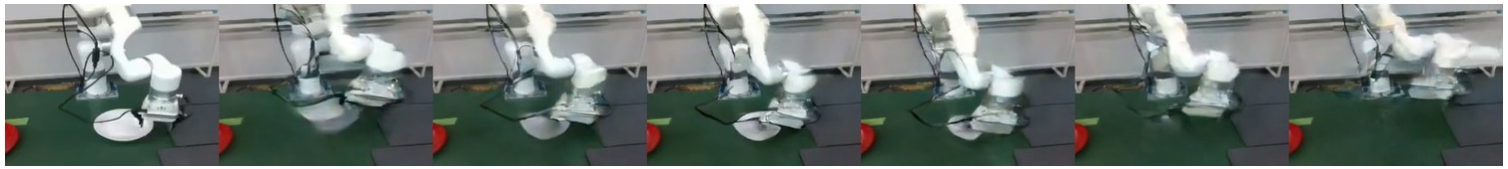
- Resolution, prediction length and steps are not matched; each model's public inference config is compared
- Speed measured at 288×384; quality evaluated at 192×256
- Drops to 7.63 TempFPS as the KV cache grows; cold start 6.41
- Counts only chunked world-model generation, excluding the policy loop
- † CausalWM is text + first-frame conditioned, not action-conditioned

Mid-trained action interface transfers: new platforms in 100–200 steps

xArm 7 • Mid-trained model (200 steps)



xArm 7 • Pre-trained backbone (2000 steps, not reached)



ViperX 300 • Mid-trained model (100 steps)

ViperX 300 • Pre-trained backbone (2000 steps, not reached)

Steps to an "action-following rollout"

Platform	Mid-training	Pre-training
xArm 7 (single-arm)	**200**	2000 †
ViperX 300 (dual-arm)	**100**	2000 †
UMI gripper (handheld)	**200**	2000 †

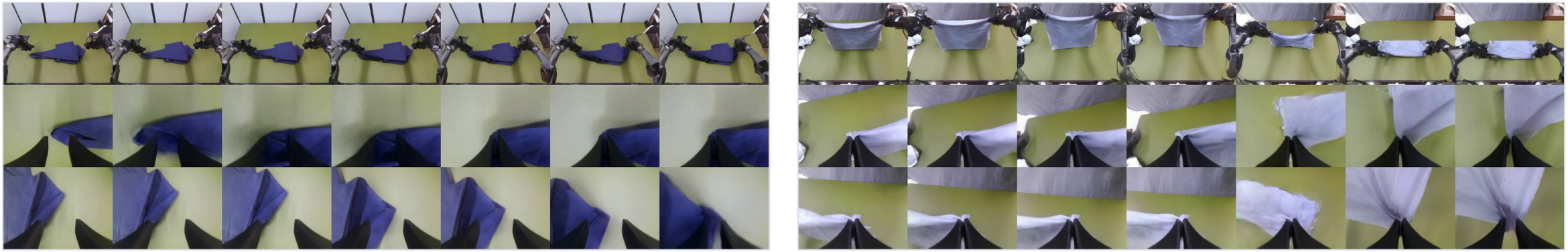
† Not reached within budget (Table 4)

- Same recipe: batch 1, lr 3e-5, no embodiment ID or dedicated parameters
- Parameters move only ~10^{-2} over 200 steps: transfer, not refitting
- Pre-trained backbone must initialize action / multi-view modules from scratch: blurry outputs, no plausible contact
- **Limitation:** cannot separate the contributions of corpus scale and of having an action interface

New scenes/cameras work zero-shot; 200 FT steps close most of the gap

(a) Zero-shot: arm control, cloth rendering; drift in rich contact

(b) FT 200 steps: arm & cloth follow commands; 3 views consistent



Rows: head / left wrist / right wrist cameras; columns: 7 frames uniformly sampled from one rollout (AgileX Piper, XVLA dataset, unseen in training)

−3.97 dB

PSNR： 19.79 → 15.82

(in-distribution XR2 → zero-shot XVLA)

+0.105

LPIPS： 0.192 → 0.297

(lower is better)

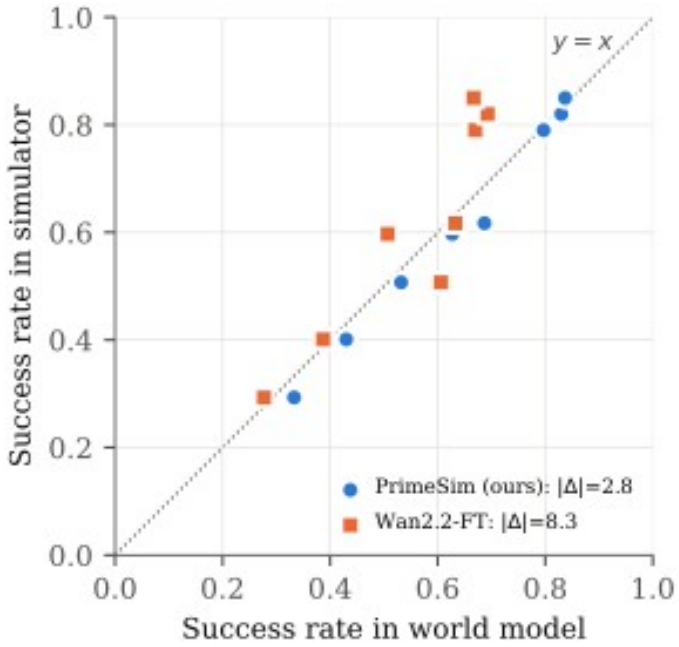
Three eval axes action controllability • object-motion plausibility • cross-view consistency

Protocol same checkpoint, 50 steps, seed 42, 192×256, 8 validation windows; SSIM not comparable across datasets

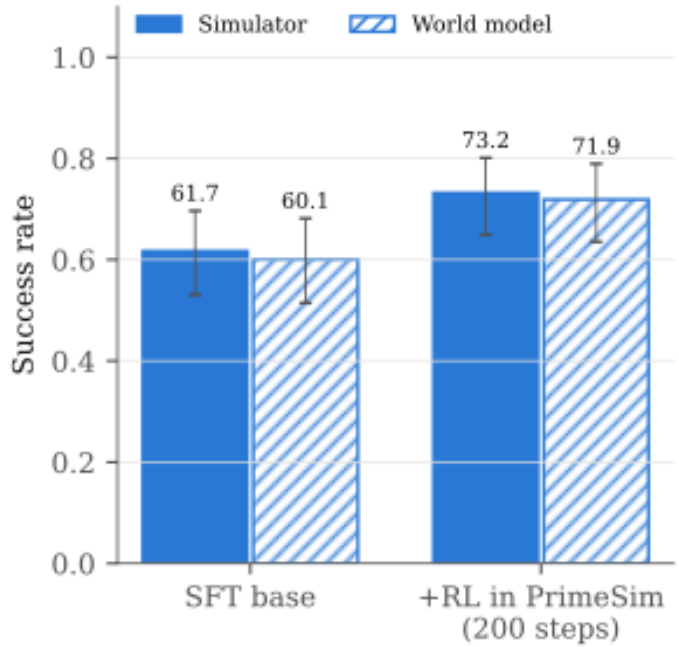
Takeaway what transfers is the "action → visual consequence" mapping, not robot appearance

Policy simulator: rankings match the simulator; RL gains transfer back

(a) Policy eval: world-model vs. LIBERO simulator success rate



(b) GRPO inside PrimeSim (group size 8), 200 steps

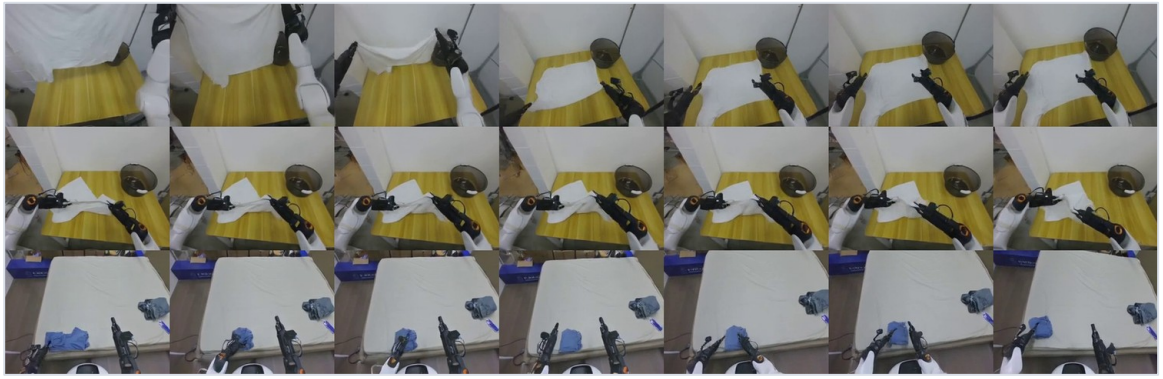


ρ=1.00	2.8 pt	18.3 pt	61.7 → 73.2%	⩽ 1.6 pt
Spearman rank	Mean abs. diff. (Wan2.2-FT 8.3)	Wan2.2-FT's largest underest. of top policy	True success rate in simulator	PrimeSim estimate − simulator

Qualitative Results

Cloth deforms smoothly; scenes stay static under counterfactual actions

Cloth manipulation (AgiBot G2 humanoid, head view, Fig. 13)



Cloth deforms continuously throughout and stays distinct from the table/bed below, never "melting into the background"

Counterfactual action following (LIBERO, Fig. 10)

Action sequence differs from the demo: the arm descends and lifts as commanded, while static elements such as the cabinet and bowl stay still

EWMBench: pick bottle from conveyor and pack it (Fig. 14 top)

EWMBench: place bread slice on plate beside toaster (Fig. 14 bottom)

Open-loop: first frame + instruction + GT action trajectory, 10 FPS, scored at 480×640

Counterfactual action following (LIBERO, Fig. 10)

Qualitative figures are only supporting evidence; for quantitative counterfactual tests see WorldSimProbe T3 (72.3, 1st)

Caveats: evidence is sim/behavior-level; several protocols need scrutiny

1 Closed loop only in sim

Policy eval and GRPO are both on LIBERO (which is in the training corpus); whether it can replace real interaction on physical robots is unverified—the authors' self-identified top limitation

2 Favorable eval protocols

WorldSimProbe adds purpose-built training data; EWMBench is evaluated after in-domain fine-tuning with sampling search; perfect HSD / nDTW / Logics scores may be saturated

3 Split speed/quality configs

Speed: 4 steps, 288×384; quality: 50 steps + CFG, 192×256; resolution, length and steps not matched across models

4 Resolution & fine control

192×256 per view; claims simulator fidelity, not pixel realism; T1 local action calibration trails LingBot-VA

5 Limited ablations

Single embodiment only, starting from pre-trained weights; action conditioning reports PSNR only; few-shot transfer measured by "steps to action following", largely qualitative

6 Textual inconsistencies

Fig. 3 states 2.8M episodes, 85:15 vs. 1.52M, 84:16 in the text / Table 12; the conclusion's "three-stage curriculum" wording is a typo

Summary

Recap: a full recipe for turning video models into robot simulators

01 Data + Interface

30,110 h of cross-embodiment robot data + unified 31-D relative action interface → transferable "action → visual consequence" mapping; new platforms in 100–200 steps

02 Lean, Effective Architecture

Frame-aligned dual-path action conditioning (mask + Adapter) + view-aware interleaved attention (camera tokens)

03 Fast Video, Slow Memory

4-step causal distillation + self-rollout asymmetric DMD + self-generated hierarchical KV memory → 10.1 FPS, stable 60–105 s rollouts

04 Simulator Value

WorldSimProbe 1st on 4/5, EWMBench 6.09; policy ranking ρ=1.00, in-model GRPO gains transfer to the simulator; next: real-robot validation

Open-source: model weights • inference code • training code | Project page: primebottech.github.io/PrimeSim-website